# Slope, Motion & Comparing Rates

**ROUND 1 OF 3** Questions 1–4

Work independently, then compare your reasoning with your group.

**Question 1** **R&P**

Position goes from 32 m at 0 s to 14 m at 6 s. Calculate average rate of change with units.

**Question 2** **R&P**

Find the slope between (-2,7) and (4,-5). Show Δy and Δx.

**Question 3** **R&P**

Two cyclists travel 30 m in 5 s and 42 m in 7 s. Compare their average rates.

**Question 4** **R&P**

A tank starts at 70 gallons and loses 4 gallons per minute. Interpret the starting value and rate.

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**ROUND 2 OF 3** Questions 5–8

Work independently, then compare your reasoning with your group.

**Question 5** **R&P**

Spiral: For {(0,5),(2,5),(2,7)}, is it a function? List domain/range.

**Question 6** **R&P**

Preview: A scooter is 12 meters from a marker at time 0 and its position increases by 3 meters each second. Write an equation for position p after t seconds.

**Question 7** **Matching practice**

An object is at 27 m after 1 s and 9 m after 7 s. Find the average rate of position change with units.

**Question 8** **Matching practice**

Find the slope through (-1,-3) and (3,9), showing Δx and Δy.

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**ROUND 3 OF 3**

Questions 9–12

Work independently, then compare your reasoning with your group.

**Question 9** **Matching practice**

Cyclist A travels 45 m in 9 s. Cyclist B travels 40 m in 8 s. Compare their average speeds.

**Question 10** **Matching practice**

A tank starts with 90 gallons and drains 6 gallons per minute. Interpret its starting value and rate of change.

**Question 11** **Matching practice**

Is {(1,2), (1,4), (3,6)} a function? State the domain and range.

**Question 12** **Matching practice**

An object starts 8 m from a marker and moves 4 m farther away each second. Write position p after t seconds.